

RAJAGANAPATHY M

AI/ML Engineer | LLM Engineer | Agentic AI | RAG | Computer Vision

✉ rajaganaa@gmail.com | 📞 +91 9176631419 | 📍 Chennai | Bangalore / Hyderabad / Remote

🌐 rajaganaa.github.io | 🌐 github.com/rajaganaa | 🌐 raja-ganapathy-36b00658 | 🌐 huggingface.co/RajGana

2+ Years Independent AI/ML Engineer (2024 – Present) | M.Tech AI (CGPA 9.6/10, SRM Institute, May 2026) | Indian Patent Filed (App No. 202641043947) | IEEE Paper — Google Scholar Indexed | 1 Live Production Product Shipped to a Paying Client (Anbu Health AI) | 3 Live HuggingFace Models | AWS + Azure + GCP | Prior 9+ years Electrical Engineering (2014–2023) | Immediate Joiner

🔧 Technical Skills

GenAI / LLMs	LLM Orchestration, RAG, FAISS, Qdrant, LangChain, HuggingFace, QLoRA, PEFT, LoRA, Agentic Workflows
Deep Learning	PyTorch, TensorFlow, Transformers, CNN, LSTM, BLIP-3, CLIP, ViT, Qwen, CodeLlama, TinyLLaMA
ML / NLP	scikit-learn, XGBoost, Gradient Boosting, SMOTE, TF-IDF, K-Means, NLTK, TextBlob, spaCy
Cloud / MLOps	AWS (SageMaker, S3, EC2), Azure (Container Apps, OpenAI), GCP (Vertex AI), Docker, GitHub Actions, FastAPI
Data Engineering	pandas, NumPy, MySQL, MongoDB, PostgreSQL, Supabase, ETL Pipelines, SQLAlchemy
Languages	Python, SQL, R, HTML, Bash

📁 Key Projects

P.1 — Anbu Health AI (Live Production Product, Paid Client) *React + FastAPI | Azure Container Apps | Groq + GPT-4o Vision | Qdrant RAG*

- Bilingual (Tamil/English) medical AI assistant analyzing lab reports, scans & medicine photos — live at anbuclinic.me, shipped to a real client under AI Vision
- 5-stage reasoning pipeline (routing → RAG → LLM → confidence scoring → hallucination check); Firebase OTP auth, DPDP Act 2023-aligned data handling

P.2 — Antahkarana Reasoning Framework (FLAGSHIP — Patent + IEEE) *Patent Filed App No. 202641043947 | IEEE Submitted*

- Cognitively-inspired 5-stage reasoning architecture for LLMs & VLMs (manas → chitta → buddhi → ahamkara → sakshi); validated on 2,500+ multimodal samples
- Powers Anbu Health AI in production | Built with Transformers, BLIP-3, Qwen, LangChain, FAISS/Qdrant

P.3 — CodeLlama Coding Assistant (Applied Fine-Tuning) *AWS SageMaker ml.g4dn.xlarge | QLoRA | PEFT | HuggingFace Spaces*

- Fine-tuned CodeLlama-7B on CodeAlpaca-20K with **60% GPU memory reduction** via 4-bit quantization
- Deployed live demo on HuggingFace Spaces with Groq API inference

🏢 Professional Experience

Founder & Independent AI/ML Engineer

Jul 2023 – Present (2+ Years)

AI Vision (Self-Employed, MSME Registered — UDYAM-TN-02-0483528)

- Designed and shipped **Anbu Health AI** end-to-end (frontend, backend, AI pipeline, cloud infra) to a paying client — live in production on Azure
- Filed 1 Indian patent, submitted 1 IEEE paper, and published 3 fine-tuned models on HuggingFace — independent, project-based work, not a salaried role

Electrical Construction Site Engineer

Jan 2017 – Jun 2023 (6.5 Years)

SR Electrical Works, Chennai

- Sole engineer across 6 landmark projects (2 hospitals, 2 hotels, 1 medical college, 1 engineering college) — 20 labourers, Rs.5–6 Cr portfolio | Zero site rejections under IEC/NBC codes
- Drove **10× profit growth** YoY by restructuring procurement & billing during GST transition (2017)

Electrical Maintenance Engineer

Dec 2014 – Dec 2016 (2 Years)

Mod Forge Pvt Ltd, Chennai (ISO/TS 16949)

- Maintained 500 kVA transformer, 250 kVA DG, motors up to 400 kW (CNC/VMC lines) — **zero unplanned downtime** over 2 years under ISO/TS 16949

🎓 Education

M.Tech — Artificial Intelligence — SRM Institute of Science & Technology, Kattankulathur

Jun 2024 – May 2026

IITM Pravartak Certified AI & ML Program — GUVI / IIT Madras + NASSCOM

Nov 2023 – Nov 2024

B.E — Electrical & Electronics Engineering — Thangavelu Engineering College, Chennai

2013

🌟 Achievements & Certifications

- **Indian Patent Filed** — App No. 202641043947 — Apr 2026
- **IEEE Paper Submitted** — Google Scholar Indexed
- **1 Product Live in Production** for a paying client (Anbu Health AI)
- **NPTEL TOP 5%** — IoT — 86% / 50,282 candidates (Oct 2025)
- **3 Live HuggingFace Models** — TinyLLaMA + Mini-VLM + CodeLlama-7B QLoRA
- **AWS Solutions Architect Associate** (Oct 2025)
- **Kaggle AI Agents Intensive** with Google (Dec 2025)
- **Hackathon 6th Place** / 195 — SRM (Dec 2024)